

How the Proof Works

We start with a smooth, incompressible flow of finite energy and keep its viscosity fixed and positive. The problem is to prove that the flow remains smooth for all time. Its velocity u and pressure p satisfy the three-dimensional Navier–Stokes equation:

$$\underbrace{\partial_t u}_{\text{acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

Strain within the fluid's flow can naturally stretch a vortex. Incompressibility narrows the material tube as it lengthens, and sustained circulation makes the narrowing flow spin faster. Viscosity spreads that rotation into the surrounding fluid, so circulation can escape the contracting tube. The vorticity $\omega = \nabla \times u$ evolves under both effects:

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The hope is simple: viscosity smooths the fluid faster than nonlinear transport can sharpen it.

If that concentration continues, faster motion occupies an ever smaller region. We can follow it toward vanishing scale to see whether viscosity eventually gains an advantage. Under the Navier–Stokes rescaling,

$$\begin{aligned} u_\lambda(x, t) &= \lambda u(\lambda x, \lambda^2 t), \\ p_\lambda(x, t) &= \lambda^2 p(\lambda x, \lambda^2 t), \quad \lambda > 1. \end{aligned}$$

The viscosity stays fixed. Transport and diffusion scale together, so the equation allows the competition to persist however small the region becomes. The velocity grows with λ , yet the volume carrying that motion shrinks fast enough for its energy to fall. Spatial derivatives reveal the concentration that energy can miss. At Sobolev order s , measuring s derivatives in L^2 ,

$$\begin{aligned} \|u_\lambda(\cdot, t)\|_{\dot{H}^s} &= \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}, \\ E[u_\lambda](t) &= \lambda^{-1} E[u](\lambda^2 t), \quad E = \frac{1}{2} \|u\|_2^2. \end{aligned}$$

Below $s = \frac{1}{2}$, the norm falls as the flow concentrates; above it, the norm grows. At $s = \frac{1}{2}$, it is invariant: the critical norm can stay fixed while speed rises without bound and energy tends to zero. Finite energy alone cannot exclude that behavior.

Nor does finite time exclude it. Each contraction by a fixed factor λ shortens the corresponding time by λ^2 . If the fluid could sustain successive contractions, infinitely many could fit before a finite time T_* :

$$\tau_j = \lambda^{-2j} \tau_0, \quad \sum_{j=0}^{\infty} \tau_j = \frac{\tau_0}{1 - \lambda^{-2}} < \infty.$$

That is the possible Zeno cascade. Rescaling leaves room for it; the regularity problem is whether the evolving fluid can actually complete it.

For a singularity to form in finite time, velocity would have to blow up. That requires acceleration to blow up, then jerk, and so on, and so on. Here these are temporal derivatives of the velocity field, with blowup understood in spatial supremum norm as $t \uparrow T_*$. The equation ties each of those temporal demands to spatial derivatives:

$$\begin{aligned}\partial_t u &= \nu \Delta u - (u \cdot \nabla)u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla)u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla)u - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p.\end{aligned}$$

Each further temporal order reaches another Laplacian into the preceding response and differentiates the nonlinear terms along with it. Expanding those responses in the original velocity exposes Δu , $\Delta^2 u$, $\Delta^3 u$, and higher spatial orders. The temporal escalation is also a demand on the spatial structure of the flow.

This becomes particularly useful at the critical Sobolev order. Write $E_s = \frac{1}{2} \|\Lambda^s u\|_2^2$, with $\Lambda = (-\Delta)^{1/2}$, and $H = E_{1/2}$. The viscous contribution to the change of E_s is $-2\nu E_{s+1}$. Starting from H , successive temporal derivatives expose successively higher spatial energies:

$$\begin{aligned}E'_s &= -2\nu E_{s+1} + \mathcal{P}_s, \\ H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= 4\nu^2 E_{5/2} - 2\nu \mathcal{P}_{3/2} + \partial_t \mathcal{P}_{1/2},\end{aligned}$$

where the nonlinear contribution is

$$\mathcal{P}_s = -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle_{L^2}.$$

The pattern continues through $E_{7/2}, E_{9/2}, \dots$. The nonlinear terms remain coupled to those rising spatial orders throughout the evolution.

Those rising spatial orders are also the orders that give smoothness. For any finite number of continuous derivatives, a sufficiently high Sobolev order is enough:

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3) \quad \text{when } m > k + \frac{3}{2},$$

and finite energy supplies the low frequencies. Taking all the orders together gives

$$L^2 \cap \bigcap_{n \geq 0} \dot{H}^{n+1/2} = \bigcap_{m \geq 0} H^m = H^\infty \hookrightarrow C^\infty.$$

The spatial requirements of smoothness have emerged from following the temporal requirements of a singularity. If the equation carries this hierarchy through T_* , the velocity reaches that time with enough regularity to continue. The question has become whether the derivatives generated by the initial flow can remain together in this intersection at the endpoint.

If a singularity were to form, we would expect smoothness at every earlier instant but loss of a fixed Sobolev norm on the approach to T_* . Velocity and its temporal responses would become unbounded even while energy remained finite. All the derivatives could exist before T_* without reaching it as derivatives of a smooth limiting velocity.

We begin the construction with the derivatives determined by the initial velocity $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$. Incompressibility fixes pressure through $-\Delta p = \partial_i \partial_j (u_i u_j)$, with its decay normalization. The differentiated equations above already show how one temporal row generates the next. Writing $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$ retains the full pattern:

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n,$$

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}.$$

Here R_i are the Riesz transforms, with repeated spatial indices summed. At $t = 0$, the recurrence starts from $V_0 = u_0$; spatial differentiation fills out each temporal row. This gives a mixed velocity and pressure jet whose finite pieces agree wherever they overlap.

Fixing a temporal depth N and spatial height M selects a finite rectangle $\mathcal{R}_{N,M}[u_0; T_*]$. The datum-generated rectangle theorem, Theorem 4.6, gives adjacent rows square integrable in H^{M+1} and H^{M-1} all the way to T_* . These finite pieces agree on overlaps and assemble into the compatible intersection:

$$\mathcal{I}[u_0] := (\mathcal{R}_{N,M}[u_0; T_*])_{N,M \geq 0},$$

$$u \in \bigcap_{r,m \geq 0} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

The intersection's adjacent temporal coordinates give

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_n \in C([0, T_*]; H^M),$$

$$\partial_t V_n = V_{n+1} \in L^2(0, T_*; H^{M-1}),$$

The Hilbert-triple trace theorem gives the last inclusion. Compatibility identifies the traces across all rectangles, leaving one endpoint in every spatial Sobolev space:

$$u_* = \lim_{t \uparrow T_*} u(t) \quad \text{in every } H^m,$$

$$u_* \in H^\infty_\sigma \hookrightarrow C^\infty.$$

Local existence starts a smooth flow from u_* , and uniqueness joins it to the original solution. The supposed maximal lifespan extends beyond T_* . A finite T_* is impossible.

We can now extract the finite estimates and use them. At any finite derivative orders and any finite time T , Sobolev embedding gives

$$\partial_t^r u \in C([0, T]; H^m) \hookrightarrow L^\infty(0, T; W^{k,p}), \quad m > k + \frac{3}{2} - \frac{3}{p},$$

for $r, k \geq 0$ and $2 \leq p \leq \infty$. In particular, the total strain accumulated in finite time is finite:

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq T \sup_{0 \leq t \leq T} \|\nabla u(t)\|_\infty < \infty.$$

Returning to a locally circular material tube, axial strain σ gives $\dot{r}/r = -\sigma/2$, with $\sigma \leq \|\nabla u\|_\infty$. Its radius obeys

$$r(t) \geq r(0) \exp\left(-\frac{1}{2} \int_0^t \|\nabla u(s)\|_\infty ds\right) > 0.$$

Only a finite amount of stretching can accumulate in a finite time. The tube cannot be squeezed to zero radius there.